

Successful Allogeneic Hematopoietic Stem Cell Transplantation for Severe Inflammatory Bowel Disease – IL10 Receptor Deficiency May Serve as a Novel Therapeutic Paradigm

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Abstract

Abstract 2379

Inflammatory bowel diseases (IBDs) comprise a heterogeneous group of disorders, classically defined as Crohn's disease, ulcerative colitis, and indeterminate colitis. The molecular pathophysiology of enterocolitis is still largely unknown. Recently, we identified monogenic mutations in the IL10 receptor genes, providing novel insights into the role of IL10-mediated immune homeostasis in the human gut.

Here, we report a series of 8 patients with mutations in the IL10RA or IL10RB gene. All patients presented within the first three months of life with severe enterocolitis and rectal fistulations. Patients were unresponsive to immunosuppressive and immunomodulatory therapies.

Deleterious mutations in *IL 10 receptor* genes abrogated IL10-induced signaling, as demonstrated by deficient STAT3 phosphorylation at the tyrosine 705 residue upon IL10 stimulation. As a consequence of defective STAT3 signal transduction in response to IL10 stimulation, IL10R-deficient mononuclear cells showed increased secretion of TNF α and various other proinflammatory cytokines unresponsive to IL10-dependent negative feedback regulation (TGF β 1, IL1 α , IL1 β , IL2, IL6, IL6sR, RANTES, MCP1, MIP1 α , and MIP1 β).

In view of the critical immunomodulatory role of IL10, we designed a protocol for allogeneic hematopoietic stem cell transplantation for patients with IL10R-mutations.

The conditioning regimen includes treosulfan, fludarabin, thiotepa, and campath-1A. A strict gut decontamination regime was chosen to reduce intestinal bacterial load. Ciclosporin A, mycophenolatemofetil, and campath-1A was used for graft versus host disease prophylaxis.

4 patients (age at HSCT 0.9–13.8yrs, 3 male, 1 female) were transplanted from a HLA-identical matched sibling (n=2) or a matched unrelated donor (MUD, HLA match 9/10) (n=2). All patients engrafted (neutrophil engraftment 10–26 days). In 3/4 patients displayed full chimerism after the first HSCT. One patient developed mixed chimerism on day 69 after HSCT and consecutively lost his graft. He was successfully re-transplanted from a different 9/10 MUD successfully achieving full donor chimerism. Acute GvHD II° and III° (skin) occurred in one patient each and was treated successfully, no chronic GvHD was observed. Regimen related toxicity was low and included mucositis I-II° and skin toxicity I-II°. Viral infections were diagnosed and successfully treated in 3/4 children (1 CMV, 1 adenovirus, 1 rotavirus). These patients responded well to the antiviral treatment or reduction of immunosuppression (follow up after HSCT 6 months-2 yrs). Intestinal inflammation resolved as proven by clinical examination, colonoscopy and histological analysis. All children improved their growth rate following the bone marrow transplantation. Post transplant samples taken after transplant showed a restored response to IL10 in mononuclear cells.

These data suggest that allogeneic HSCT may represent a novel, effective, and safe therapeutic approach to treat defined subgroups of IBD patients.

Disclosures:

No relevant conflicts of interest to declare.

Author notes

* Asterisk with author names denotes non-ASH members.

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